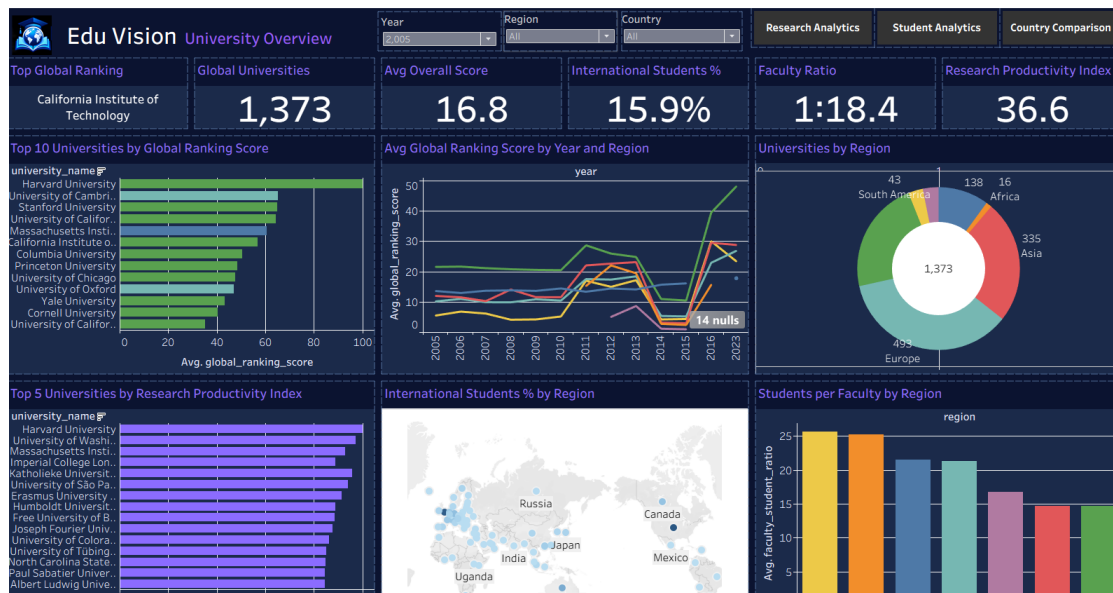


EDUVISION

Dashboard Development Process

Higher Education Performance Dashboard | Tableau



Project objective

EduVision brings university performance indicators into one interactive Tableau workbook. The four dashboards shown in the supplied screenshots cover University Overview, Research Analytics, Student Analytics, and Country Comparison.

Scope of this document

This six-page development record describes the visible KPI cards, charts, filters, navigation, and quality checks. It is based on the supplied screenshots; it does not claim that unseen calculations, relationships, or interactions have been tested.

1. Dashboard Development Workflow

The development process moves from prepared data to Tableau worksheets, dashboard assembly, interaction testing, and final packaging.

Step 1 — Prepare the analytical dataset

Earlier project milestones cover university data collection, cleaning/transformation, and KPI engineering. The dashboards visibly use fields such as `university_name`, `country`, `region`, `year`, `global_ranking_score`, `research_productivity_index`, `research_impact_score`, `international_student_pct`, and `faculty_student_ratio`.

Step 2 — Connect and validate data in Tableau

Connect Tableau to the prepared dataset. Confirm that university name, country, region, and year are treated as dimensions, while ranking, research, international-student, and faculty-ratio fields are assigned suitable numeric types. Review missing values before interpreting results.

Step 3 — Build worksheet-level visuals

- Create KPI sheets for university count, average score, international-student percentage, enrollment, and faculty ratio.
- Create ranked bars, trend lines, regional distribution charts, scatter plots, maps, treemaps, and comparison tables.
- Use the correct dimension to group each measure and choose aggregations that match the question being answered.

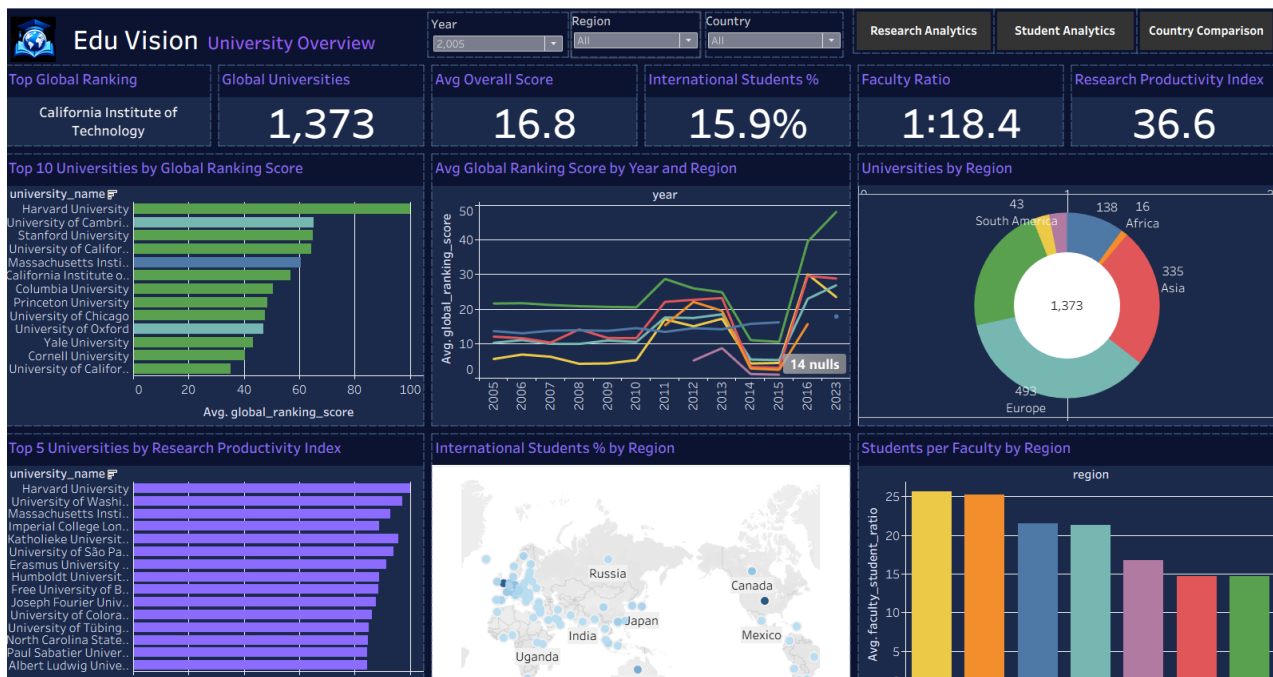
Step 4 — Assemble dashboard pages

Arrange related worksheets into four pages. The screenshots show a consistent dark navy theme, purple headings, top-level Year/Region/Country filters, and navigation buttons linking the dashboard pages.

Step 5 — Test and package

Check filter behavior, chart titles and units, null/unknown records, crowded labels, and any external visualization extension. Save the final workbook as a packaged Tableau workbook (.twbx) and reopen it to verify the deliverable.

2. Dashboard 1 — University Overview



Purpose

The Overview page gives a high-level summary of university performance and allows the audience to move from headline indicators to institution rankings, regional patterns, and research/student measures.

Visible components

- Top Global Ranking card identifies the leading institution displayed in the current view.
- KPI cards show Global Universities, Avg Overall Score, International Students %, Faculty Ratio, and Research Productivity Index.
- Top 10 Universities by Global Ranking Score compares institutions with horizontal bars.
- Avg Global Ranking Score by Year and Region shows trends across time and regions.
- Universities by Region uses a donut chart to show regional distribution.
- Top 5 Universities by Research Productivity Index ranks institutions by the displayed productivity measure.
- International Students % by Region provides a geographic view.
- Students per Faculty by Region compares regional faculty/student patterns.

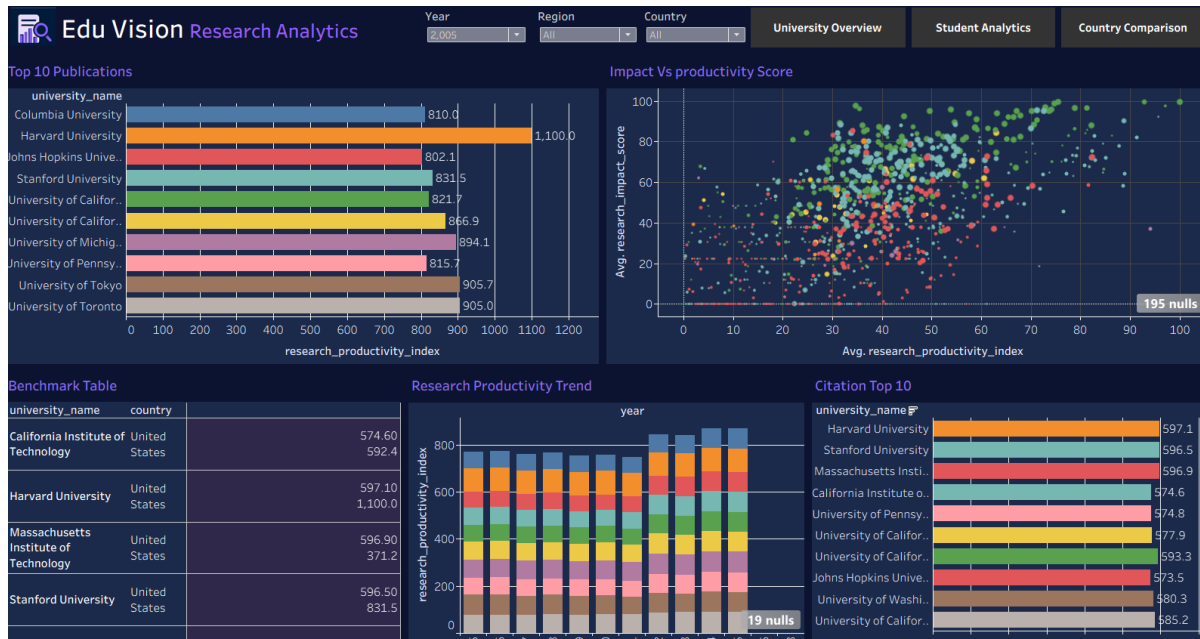
Presentation wording

"This page is the starting point for the analysis. The KPI cards summarize the dataset, ranked bars compare universities, the line chart shows changes across years and regions, and the regional charts add geographic context. The Year, Region, and Country filters help narrow the view."

QA note

The screenshot shows 14 nulls on the ranking trend and the year displayed as '2,005'. Confirm the Year field's data type and number format, and inspect records missing ranking scores before drawing conclusions.

3. Dashboard 2 — Research Analytics



Purpose

Research Analytics focuses on productivity and impact measures. It compares institutions, explores the relationship between productivity and impact, and provides a benchmark table and a time-based trend.

Visible components

- Top 10 Publications is a horizontal bar chart; its axis is labelled `research_productivity_index`.
- Impact Vs Productivity Score is a scatter plot with average research productivity index on the horizontal axis and average research impact score on the vertical axis.
- Benchmark Table displays university and country information alongside research-related values.
- Research Productivity Trend shows the productivity index by year using stacked bars.
- Citation Top 10 compares institutions by the displayed citation-related measure.
- Shared Year, Region, and Country filters and navigation buttons connect this page to the other dashboards.

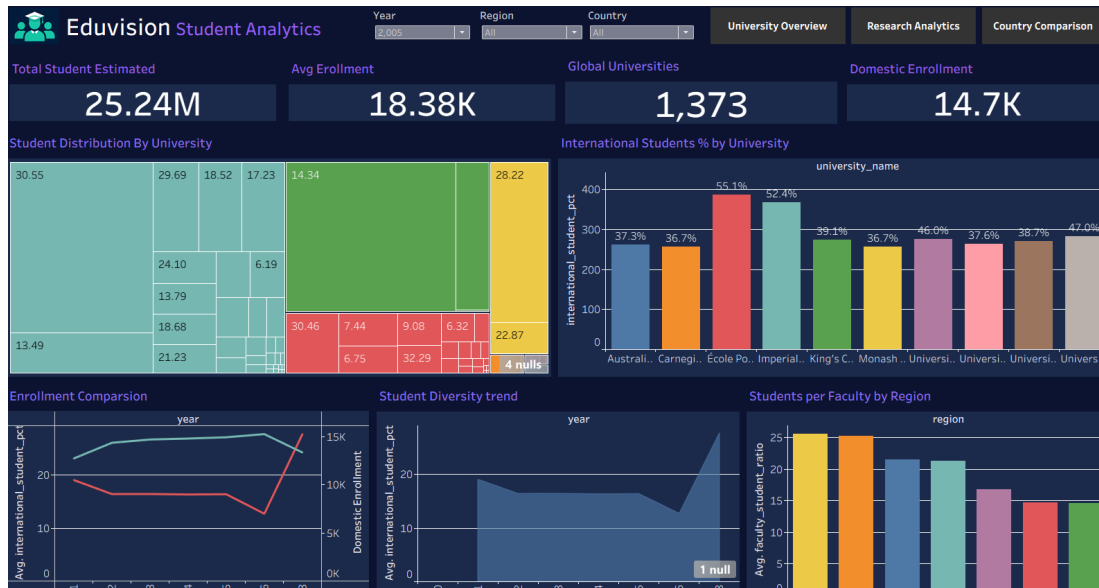
Presentation wording

"This page compares research productivity and impact. Ranked bars highlight institutions with higher displayed values. The scatter plot helps inspect how productivity and impact vary together, while the trend chart shows the productivity measure across years. The benchmark table supports institution-level review."

QA note

The scatter plot shows 195 nulls and the productivity trend shows 19 nulls. Some displayed values are decimal numbers although the chart title refers to publications. Verify the underlying fields, units, aggregation, and missing-value treatment.

4. Dashboard 3 — Student Analytics



Purpose and visible components

Student Analytics summarizes student population, enrollment, international participation, and faculty capacity. Visible KPI cards include Total Student Estimated, Avg Enrollment, Global Universities, and Domestic Enrollment. Other views include a student-distribution treemap, international-student percentage by university, enrollment comparison, student-diversity trend, and students-per-faculty by region.

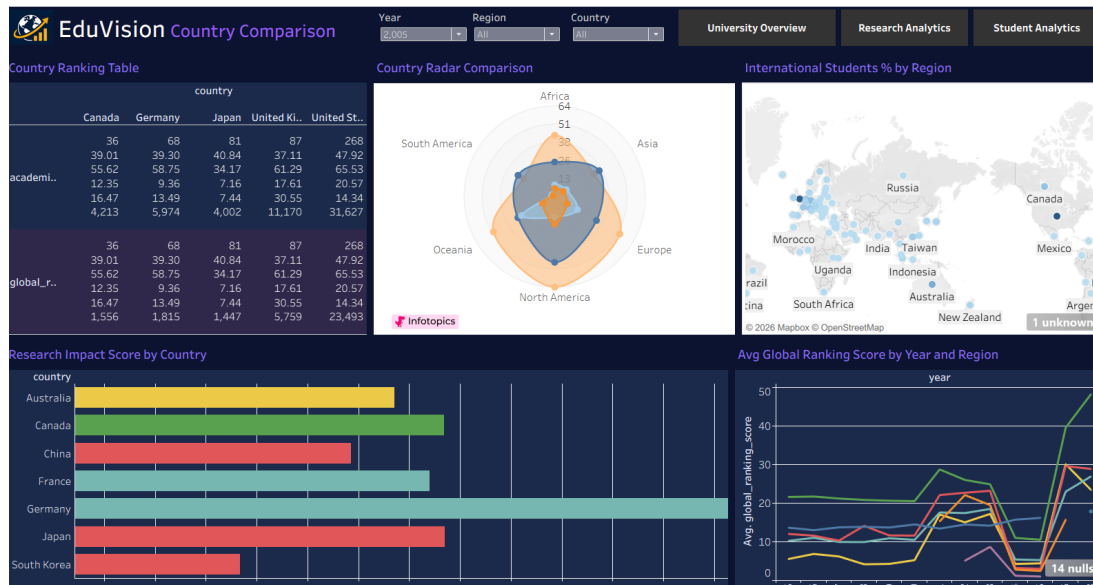
Presentation wording

"This dashboard focuses on student scale and participation. KPI cards summarize enrollment-related indicators, the treemap shows distribution across universities, the trend charts show changes over time, and the regional ratio chart compares students per faculty."

QA note

Null indicators appear in the treemap and diversity trend. The international-student percentage bars show percentage labels, while the axis scale appears to extend into the hundreds. Verify the measure and aggregation used on the axis and labels so that percentages are interpreted consistently.

5. Dashboard 4 — Country Comparison



Purpose and visible components

Country Comparison shifts the analysis from individual institutions to country and regional comparisons. It includes a Country Ranking Table, Country Radar Comparison, International Students % by Region map, Research Impact Score by Country bar chart, and ranking-score trend by year and region.

Presentation wording

“This page compares higher-education indicators across countries and regions. The table provides country-level values, the radar chart compares multiple dimensions, the map adds geographic context, and the bar chart compares research-impact values by country.”

QA note

The radar visualization shows an Infotopics extension mark, so note this dependency when sharing the workbook. The map shows 1 unknown; review the geographic role or spelling of the country value that Tableau cannot locate.

6. Integration and Final QA

- Confirm Year, Region, and Country filters behave as intended on all four dashboards.
- Test navigation buttons and any filter/highlight actions.
- Verify titles, axes, units, and aggregations against the fields actually used.
- Investigate visible Null/unknown indicators; do not silently replace missing values with zero.
- Check long institution/country labels for truncation and overlap.
- Confirm the radar extension is available in the demonstration environment.
- Save and reopen the final .twbx to check that the workbook is portable.

Repository and conclusion

Keep the packaged workbook, dashboard screenshots, and this PDF in the appropriate GitHub project folders. Reviewers can download the .twbx and open it in compatible Tableau software. For browser-based access, publish separately and check visibility and data-privacy settings. EduVision combines four complementary views—university overview, research analytics, student analytics, and country comparison—into one performance dashboard workbook.